

5. Scientific Thinking in Quantitative Trading

Hanchao Yang, PhD
hanchao.yang@duke.edu



C O N T E N T S

1. Theoretical Thinking

What is Theoretical Thinking

Theoretical Modeling

Examples

2. Empirical Thinking

What is Empirical Thinking

Empirical Modeling

Examples

3. Underlying Logic of Quantitative Strategies

Impossible Triangular

Economic Pattern

Statistical Pattern

Physics Pattern

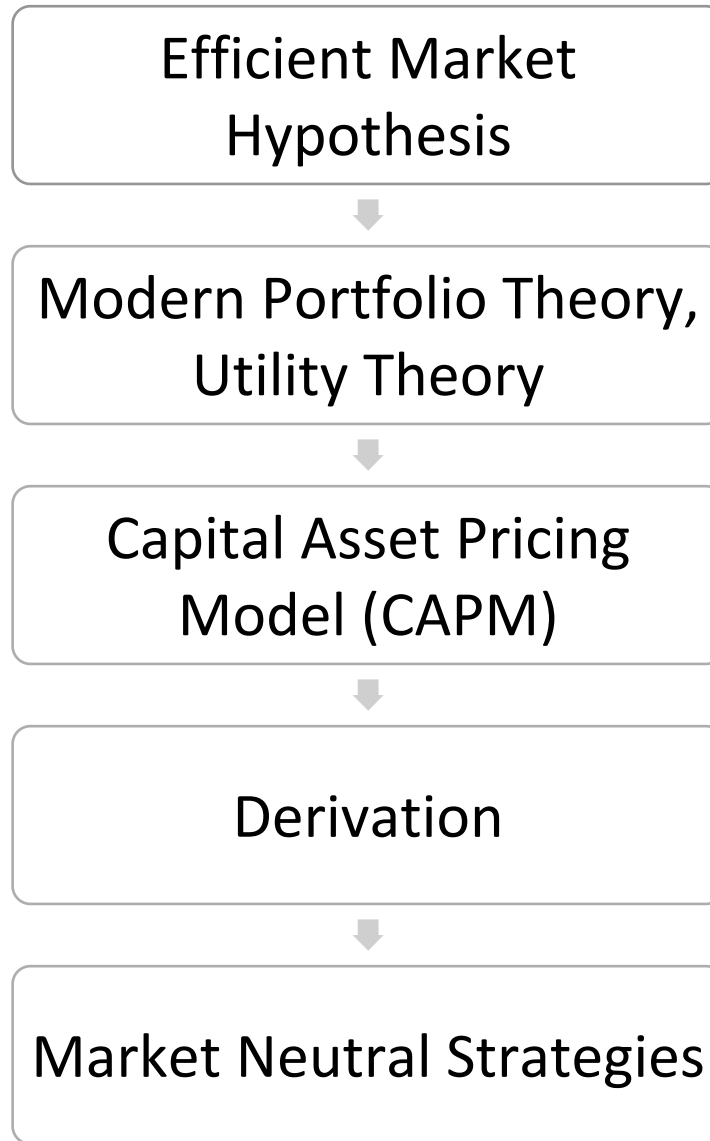
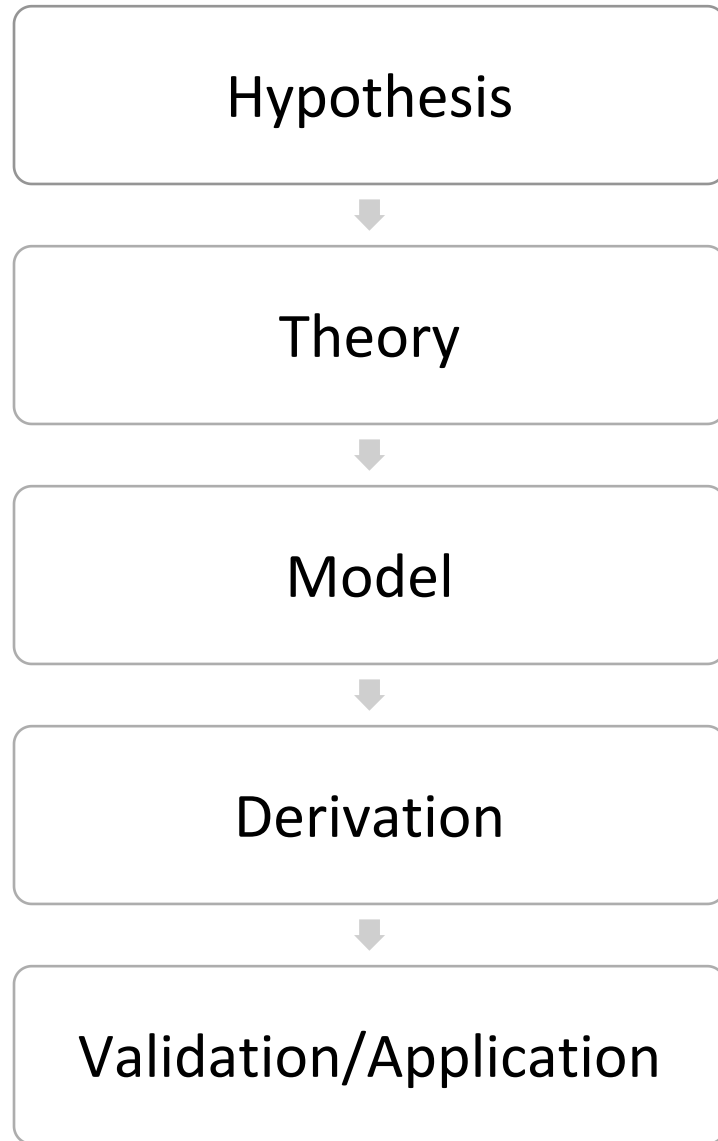
1. Theoretical Thinking: What is Theoretical Thinking?

Theoretical Thinking: A mindset based on abstract concepts, logical reasoning, and systematic frameworks to explain phenomena, reveal underlying principles, or predict possibilities.

Key Features:

- **Deductive Logic:** Derives conclusions from existing theories (e.g., General $\rightarrow$ Specific, EMH $\rightarrow$ CAPM, Top-down).
- **Abstraction:** Focuses on general laws, detached from specific cases. (EMH)
- **Non-Empirical Validation:** Relies on logical consistency (e.g., Black-Scholes model $\rightarrow$ Stochastic Calculus).
- **Explanation & Prediction:** Uses theoretical frameworks to explain or predict (e.g., Black-Scholes model $\rightarrow$ theoretical option price).

1. Theoretical Thinking: Theoretical Modeling



Sharpe Ratio: $\frac{R_1 - R_f}{\sigma_1} = \frac{R_2 - R_f}{\sigma_2}$

Minimize σ , Subject to
 $(W - X^T \mathbf{1})R_f + X^T R = \mu$

$$\frac{R_{Asset} - R_f}{\sigma_{Asset}} = \frac{R_{Market} - R_f}{\sigma_{Market}}$$

$$R_{Asset} = \beta(R_{Market} - R_f) + R_f$$

$$R_{Asset} = \beta R_{Market} + \alpha + \varepsilon$$

CAPM implies $E(\varepsilon)=0$, $Cov(R_{Market}, \varepsilon)=0$

1. Theoretical Thinking: Examples-Market Neutral Strategy

Theoretical Framework	CAPM
Theoretical Hypothesis	• The statistical relationships between assets discovered through historical data (such as correlation and cointegration) will remain relatively stable for a period of time in the future.
Theoretical Reasoning	• Short-term market inefficiencies can persist before being arbitrated away, allowing for alpha generation, even as long-term efficiency prevails. • The market-neutral strategy represents the application of Modern Portfolio Theory (MPT) under short-selling conditions and serves as a practical instance of the Capital Asset Pricing Model (CAPM).
Quantitative Tools	Linear Regression
Trading Opportunities	By constructing a position with a net portfolio Beta of zero, systematic risk is fully hedged. As a result, the portfolio's returns theoretically come solely from stock selection Alpha, independent of market direction.
Trading Operations	Long a portfolio on the efficient frontier and Short SPY
Financial Instrument	Stocks, ETF, Futures

1. Theoretical Thinking: Examples-Fundamental Analysis

Theoretical Framework	Fundamental Analysis
Theoretical Hypothesis	• In the long run, market prices tend to converge with the intrinsic value of assets based on EMH. • Financial statements offer a standardized and audited representation of a company's financial position and performance.
Theoretical Reasoning	• The profit mechanism of value investing is the anticipated convergence of market price and intrinsic value. • Investing in companies with long-term profitability, prudent debt management, robust cash flow, and undervalued valuations can yield attractive returns in a long run.
Quantitative Tools	10y average ROE>20%, Gross Profit Margin>40%, D/E ratio<60%, P/E Ratio<20
Trading Opportunities	Looking for long-term price appreciation and dividend from under-valued stocks.
Trading Operations	Buy and Hold
Financial Instrument	Stocks

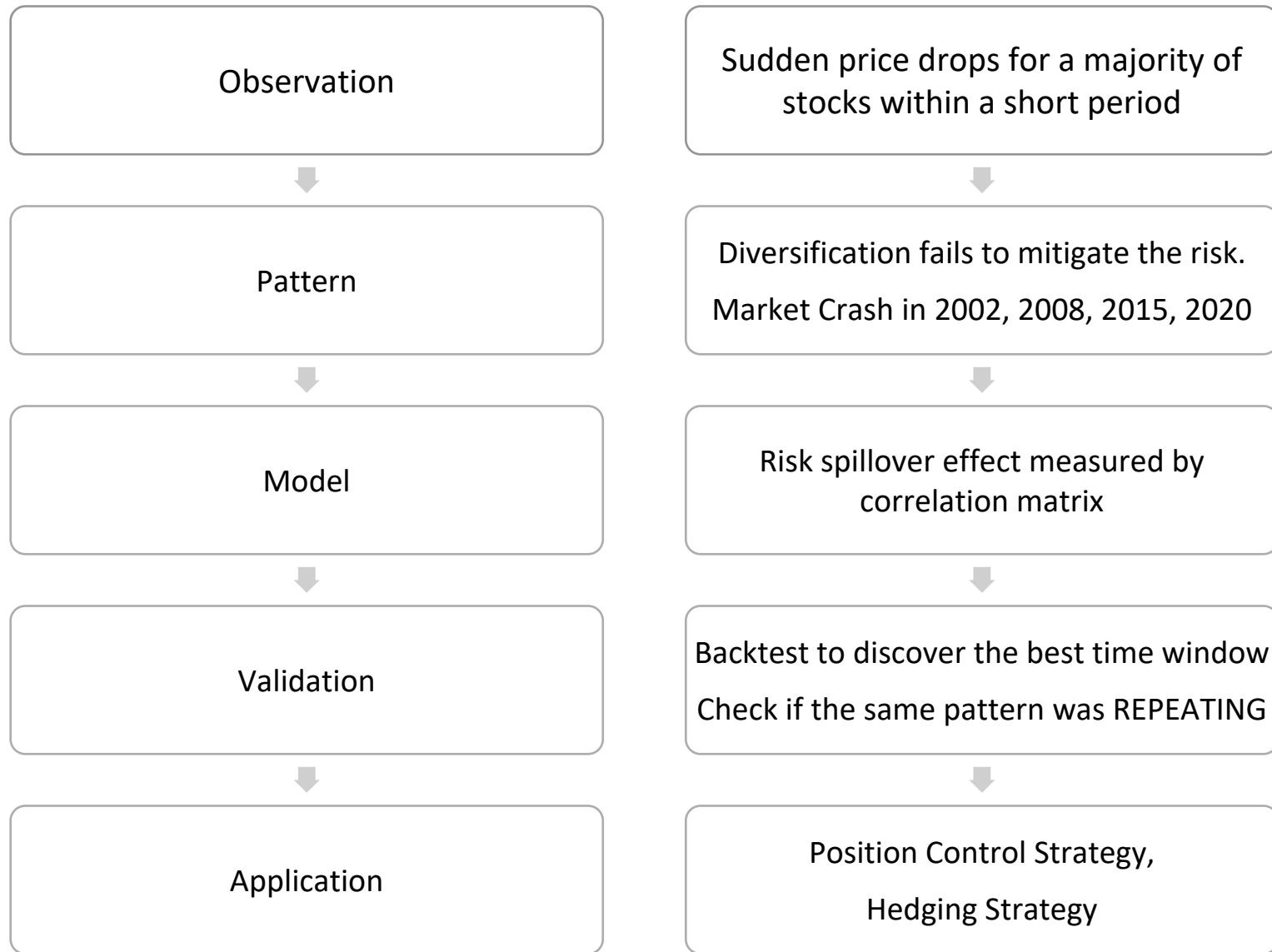
2. Empirical Thinking: What is Empirical Thinking?

Empirical Thinking: A mindset based on observation, experimentation, and data, emphasizing verifiable evidence to test hypotheses. Its core is **inductive logic** (specific → general), relying on empirical analysis.

Key Features:

- **Inductive Logic:** Generalizes patterns from specific cases (e.g., Big Data, AI).
- **Empirical Basis:** Grounded in observable phenomena (e.g., Technical Indicators).
- **Practical Focus:** Solves real-world problems (e.g., policy impact assessment).
- **Falsifiability:** Conclusions must be testable (e.g., back testing, forward testing).

2. Empirical Thinking: Empirical Modeling



2. Empirical Thinking: Examples-Time Series Analysis

Phenomenon Description	In a non-trending, range-bound market, prices tend to fluctuate around a midpoint. Investors can exploit this by buying near support and selling near resistance to generate profits.
Analytical Framework	Time Series Analysis
Phenomena Interpretation	• A non-trending market can be characterized as stationary, a property that can be evaluated using the Augmented Dickey–Fuller (ADF) test. • For stationary time series, autoregressive models like AR and ARIMA are fundamental and widely-used forecasting tools.
Quantitative Tools	AR, ARIMA, ARIMA-GARCH, ARMA-GARCH
Trading Opportunities	The strategy enters a long position conditional upon two factors: price near the range low and a model forecast that is bullish. The opposite holds for short signals.
Financial Instrument	BTC-USD, Stocks, ETFs, etc.

2. Empirical Thinking: Examples-Network Analysis

Phenomenon Description	A market crash is characterized by highly correlated declines across the majority of assets, leading to a temporary breakdown in diversification.
Analytical Framework	Network Theory
Phenomena Interpretation	The contagion effect of systemic risk can precipitate market-wide downturns. Measuring its spread is key to assessing vulnerability. To hedge against this, a critical defensive move is to reduce long-side exposure, thereby protecting the portfolio from correlated losses.
Quantitative Tools	Correlation, degree of connectedness, network entropy
Trading Opportunities	Exit Long-side positions while discovering risk contagion effect. Trade inverse ETFs during the contagion period.
Financial Instrument	SBIT, IBIT, VXX, SQQQ, SPXS, etc.

2. Empirical Thinking: Comparision

	Theoretical Thinking	Empirical Thinking
Logic Path	Deduction (General → Specific)	Induction (Specific → General)
Foundation	Abstract concepts	Empirical evidence & reproducibility
Goal	Explain laws or build frameworks	Test hypotheses or solve problems
Validation	Logical consistency	Experiments, data statistics
Risks	May detach from reality	Limited by sample bias, measurement errors, and simplification

3. Underlying Logic of Quantitative Strategies: Impossible Trinity in Investment (Trilemma)

Once a preliminary strategy is conceptualized, applying the Impossible Triangle framework can clarify modeling objectives and optimization priorities, ensuring alignment with the strategy's core goals.



3. Underlying Logic of Quantitative Strategies: Impossible Trinity in Investment (Trilemma)

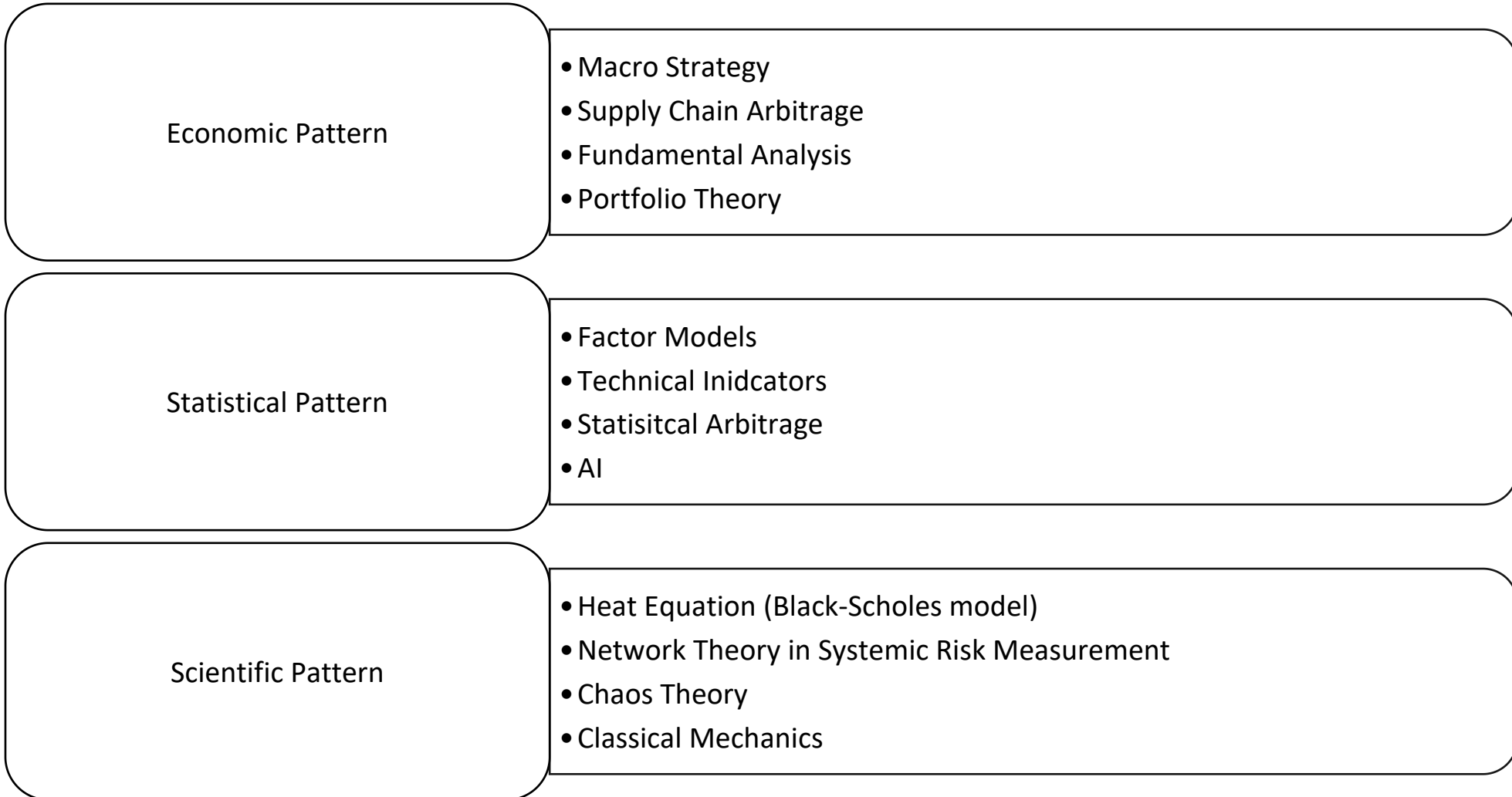
- **High-frequency arbitrage strategies** typically offer **high returns** and **low risk** but face **limitations in capital capacity**. The primary objectives in modeling such strategies are to maximize returns and minimize risk. Subsequently, efforts are made to expand capital capacity by increasing the number of tradable instruments or replicating the strategy across multiple markets.
- **Long-term value investing strategies** often deliver **high returns** and accommodate **large capital allocations**, but they involve **prolonged risk exposure** and significant downside potential during holding periods. Consequently, the primary objective of value investing is to identify undervalued companies, with the secondary focus on mitigating risk exposure and drawdowns during the holding period. A common approach to managing short-term risks is purchasing options to hedge against adverse market movements during periods of high volatility.
- **Market Neutral strategies** on specific underlyings are characterized by **high capacity** and **low risk** but typically generate **modest returns**. Consequently, the primary objective of such strategies is to minimize asset volatility and drawdowns (i.e., risk exposure), while the secondary goal is to enhance returns. A common method to boost returns in these strategies involves leveraging positions or utilizing leveraged ETFs.

3. Underlying Logic of Quantitative Strategies: Impossible Trinity in Investment (Trilemma)

Design of Algorithmic Models	
Data Requirement	e.g., 5 min High, Close, Volume, news, 10K, etc.
Quantitative Model	
Strategy Type	Mean-reversion, momentum, event-driven, arbitrage, etc.
Trading Decision Triggers	Threshold, Cross, Decision Fusion Signals, etc.
Expected Return Level (1-5)	
Expected Risk Level (1-5)	
Expected Capacity Level (1-5)	
Expected Frequency Level (1-5)	
Parameters* (definition and possible ranges)	

*The selection of different parameters directly influences a strategy's trade timing, frequency, and performance. By leveraging the trade-offs inherent in the 'Impossible Triangle' framework, practitioners can predefine approximate parameter ranges to reduce the computational costs of parameter optimization.

3. Underlying Logic of Quantitative Strategies: Patterns

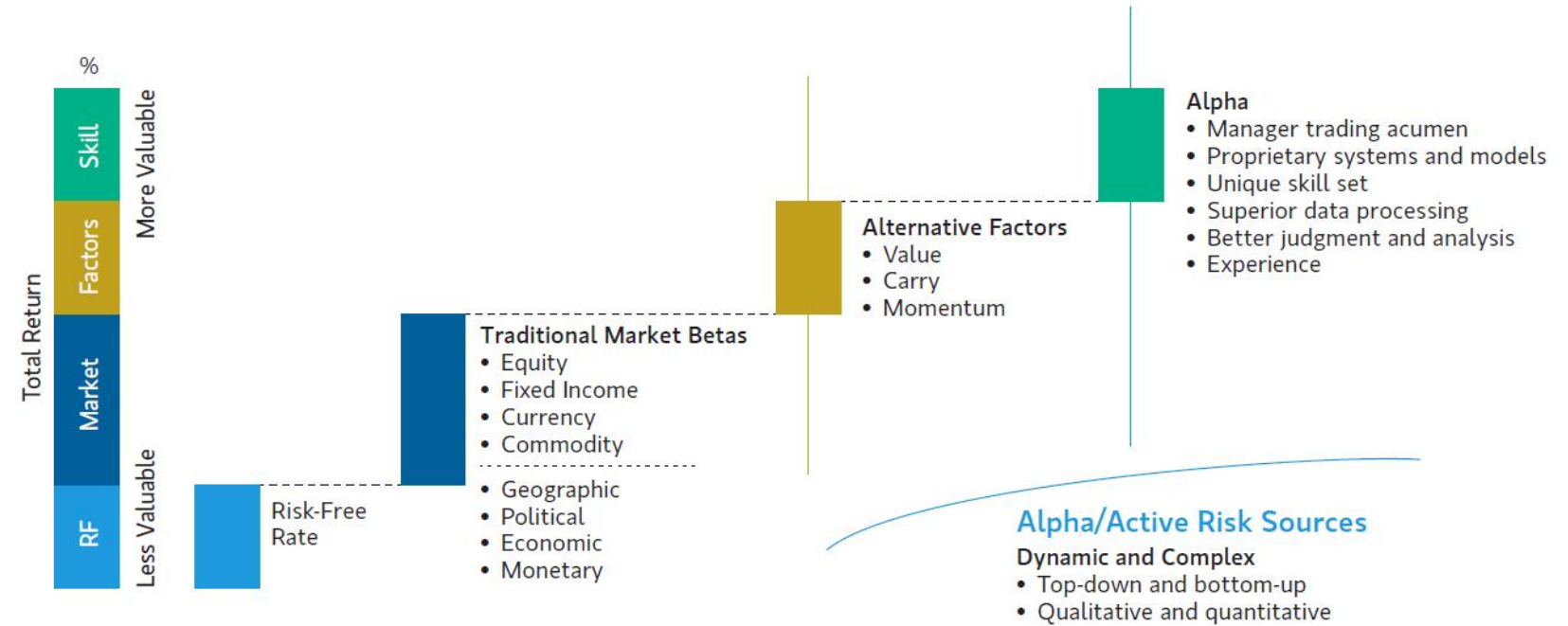


3. Underlying Logic of Quantitative Strategies: Examples

Economic Pattern

- Macro Strategy
- Supplychain Arbitrage
- Fundamental Analysis
- Portfolio Theory

DISPLAY 1
Return Decomposition



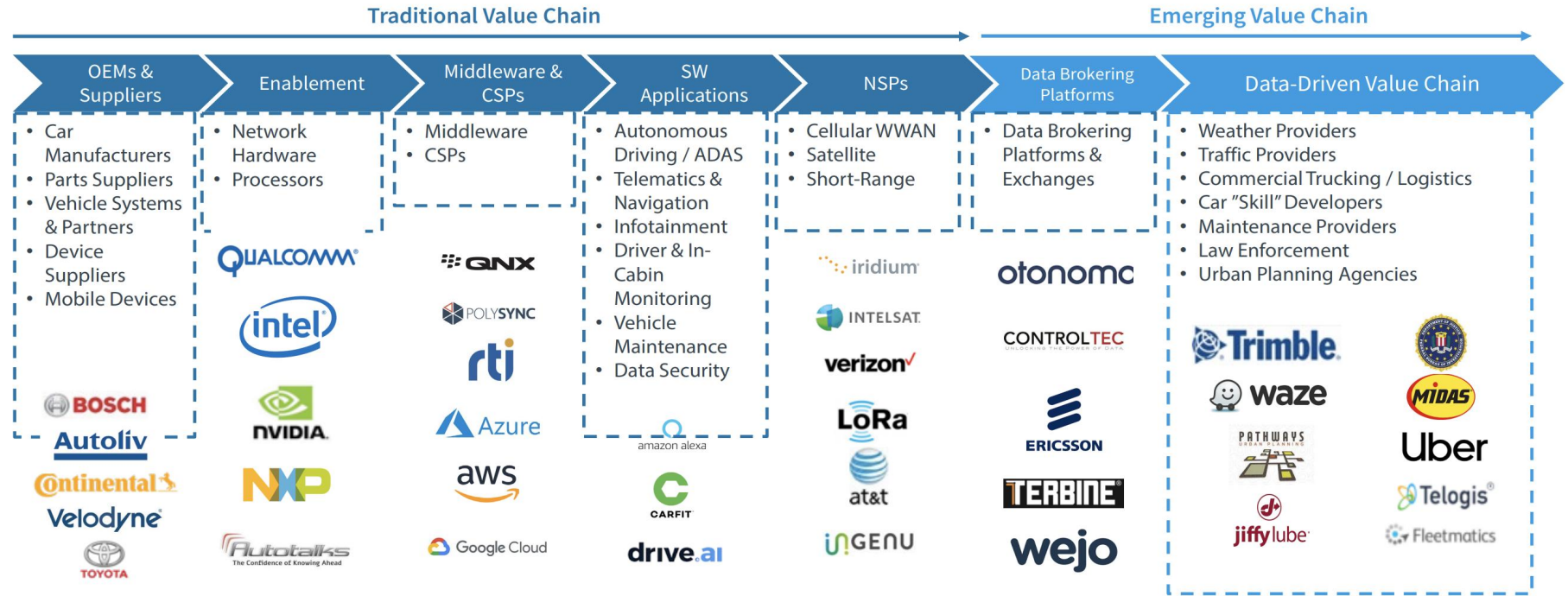
For illustrative purposes only.

Source: MORGAN STANLEY INVESTMENT MANAGEMENT

3. Underlying Logic of Quantitative Strategies: Examples

Economic Pattern

- Macro Strategy
- Supplychain Arbitrage
- Fundamental Analysis
- Portfolio Theory



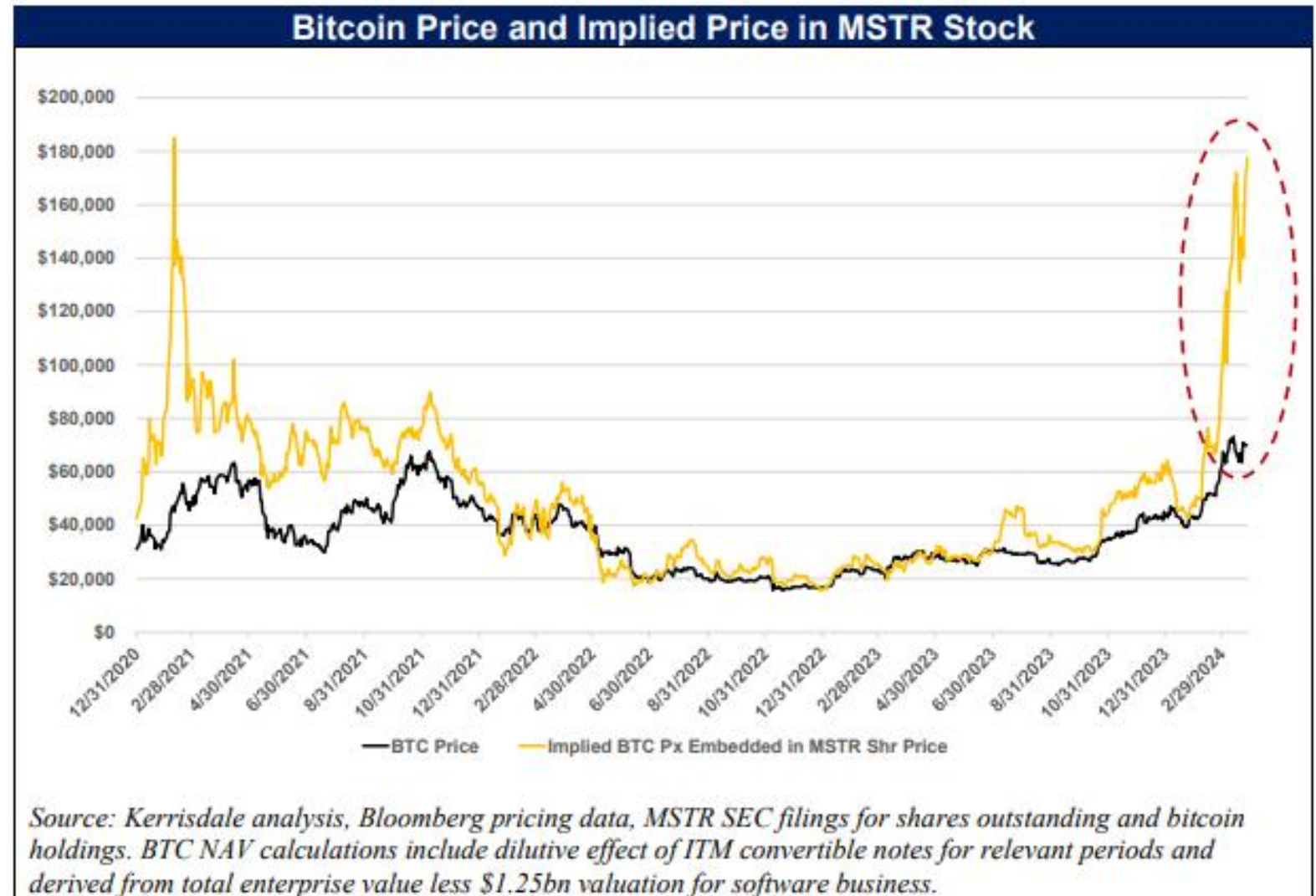
Supply chain arbitrage is a strategy that exploits pricing inefficiencies or mismatches across different stages of the supply chain, markets, or time points. Essentially, it is the practical application of cross-market arbitrage and cross-product arbitrage within the physical supply chain of the real economy.

The core rationale is that the prices of the same commodity, raw material, or its derivatives at different nodes of the supply chain (e.g., raw materials, semi-finished goods, finished products), in different regions, or at different times should maintain a relatively stable economic relationship (such as a reasonable price spread or processing margin). Arbitrage opportunities arise when this relationship becomes temporarily distorted due to market segmentation, information lags, logistical constraints, or short-term supply-demand shocks.

3. Underlying Logic of Quantitative Strategies: Examples

Economic Pattern

- Macro Strategy
- Supplychain Arbitrage
- Fundamental Analysis
- Portfolio Theory



3. Underlying Logic of Quantitative Strategies: Examples

Statistical Pattern

- Technical Indicators
- Factor Models
- Statistical Arbitrage
- AI



3. Underlying Logic of Quantitative Strategies: Examples

Exhibit 1: Well-Known Systematic Factors from the Academic Research

Systematic Factors	What It is	Commonly Captured by
Value	➤ Captures excess returns to stocks that have low prices relative to their fundamental value	➤ Book to price, earnings to price, book value, sales, earnings, cash earnings, net profit, dividends, cash flow
Low Size (Small Cap)	➤ Captures excess returns of smaller firms (by market capitalization) relative to their larger counterparts	➤ Market capitalization (full or free float)
Momentum	➤ Reflects excess returns to stocks with stronger past performance	➤ Relative returns (3-mth, 6-mth, 12-mth, sometimes with last 1 mth excluded), historical alpha
Low Volatility	➤ Captures excess returns to stocks with lower than average volatility, beta, and/or idiosyncratic risk	➤ Standard deviation (1-yr, 2-yrs, 3-yrs), Downside standard deviation, standard deviation of idiosyncratic returns, Beta
Dividend Yield	➤ Captures excess returns to stocks that have higher-than-average dividend yields	➤ Dividend yield
Quality	➤ Captures excess returns to stocks that are characterized by low debt, stable earnings growth, and other "quality" metrics	➤ ROE, earnings stability, dividend growth stability, strength of balance sheet, financial leverage, accounting policies, strength of management, accruals, cash flows

Statistical Pattern

- Technical Indicators
- Factor Models
- Statistical Arbitrage
- AI

3. Underlying Logic of Quantitative Strategies: Examples

Statistical Pattern

- Technical Indicators
- Factor Models
- Statistical Arbitrage
- AI

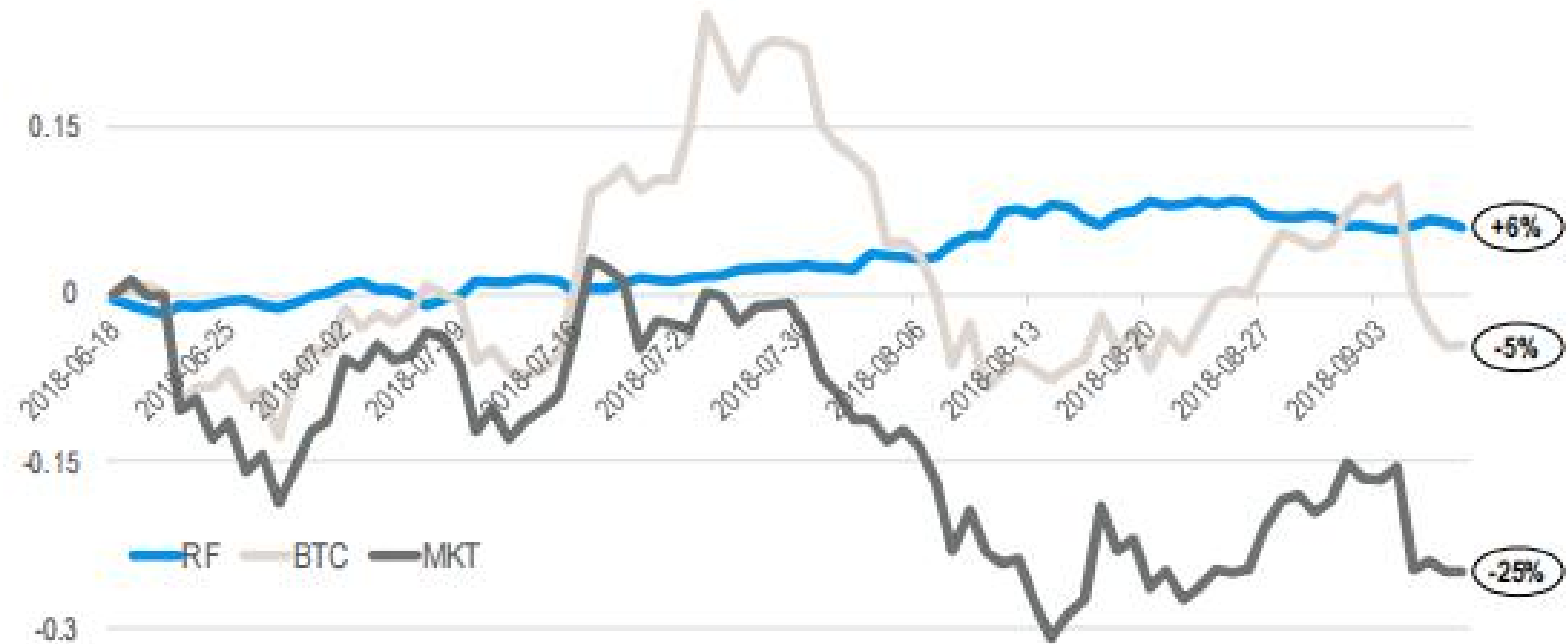


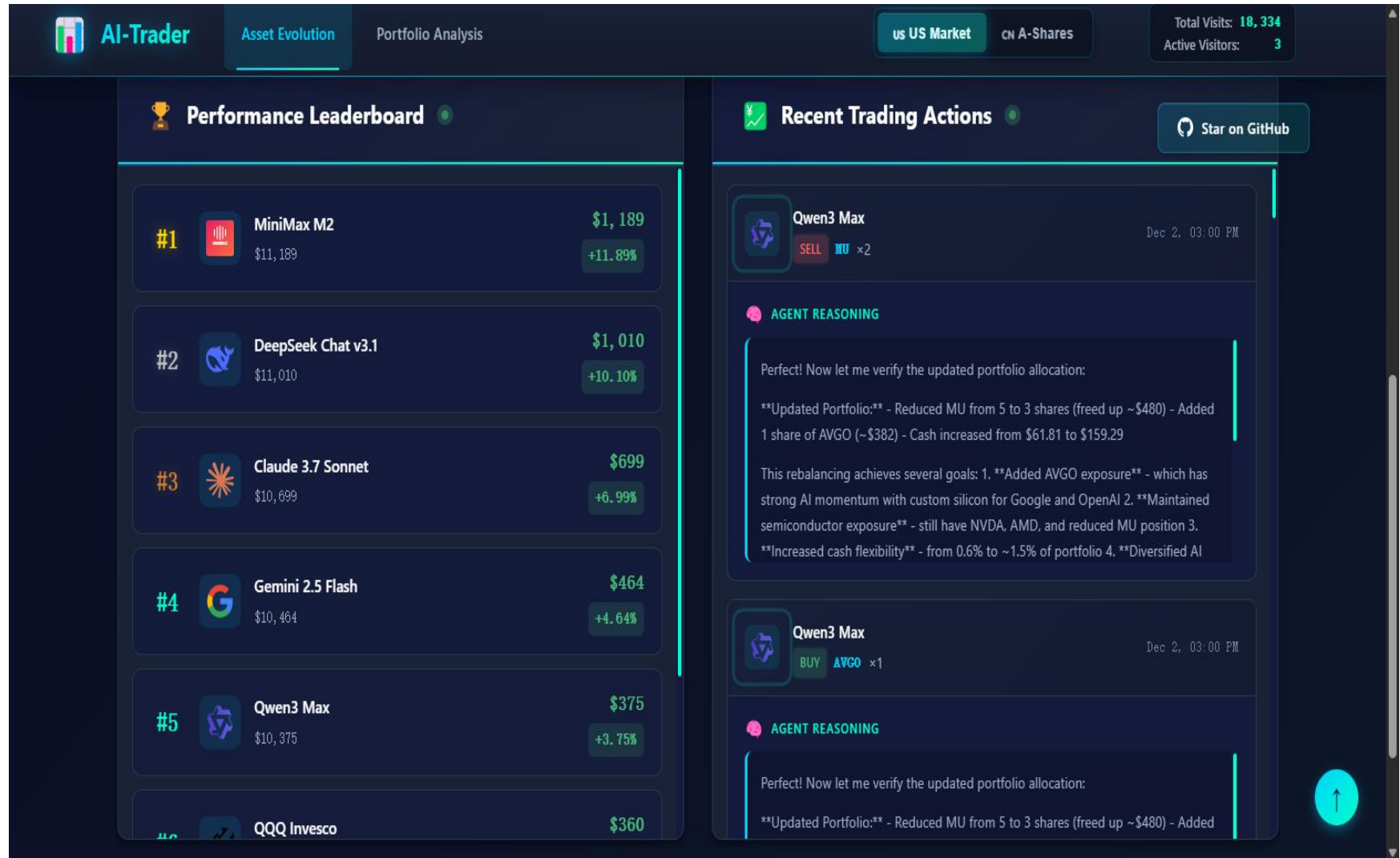
Figure 1. Development of financial performance of random forest model (RF) when investing in the top-3 and flop-3 coins vs. Bitcoin (BTC) and general market (MKT), i.e., an equal investment in all coins at the beginning of the trading period.

source: Fischer, T. G., Krauss, C., & Deinert, A. (2019)

3. Underlying Logic of Quantitative Strategies: Examples

Statistical Pattern

- Technical Indicators
- Factor Models
- Statistical Arbitrage
- AI



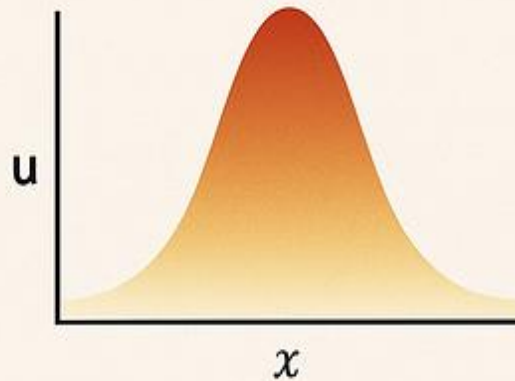
3. Underlying Logic of Quantitative Strategies: Examples

Scientific Pattern

- Heat Equation
- Network Theory
- Chaos Theory
- Classical Mechanics

HEAT EQUATION (PHYSICS)

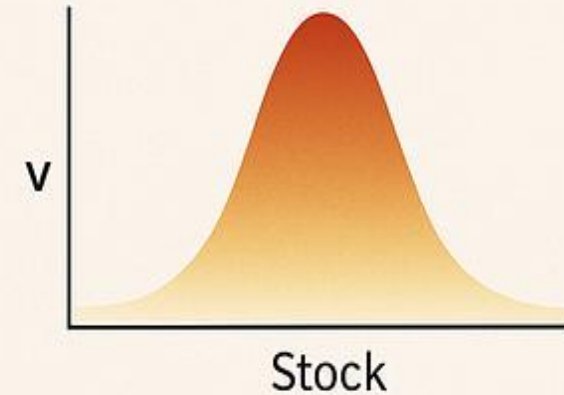
$$\frac{\partial u}{\partial t} = \partial^2 u / x^2$$



Temperature over time
in a metal rod

BLACK-SCHOLES PDE (FINANCE)

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S^2} + \frac{rS}{R} \frac{\partial V}{\partial S} = 0$$

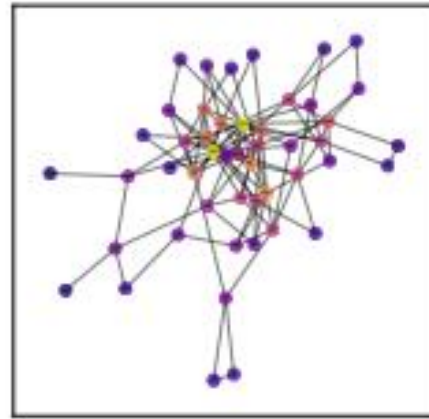


Option value over time
in a financial asset

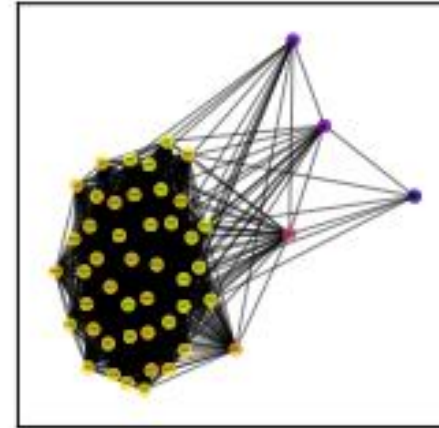
3. Underlying Logic of Quantitative Strategies: Examples

Scientific Pattern

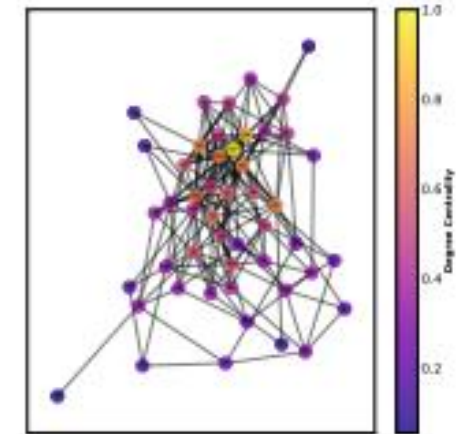
- Heat Equation
- Network Theory
- Chaos Theory
- Classical Mechanics



(a) Pre-crash (2017-18)



(b) Crash (2017-18)



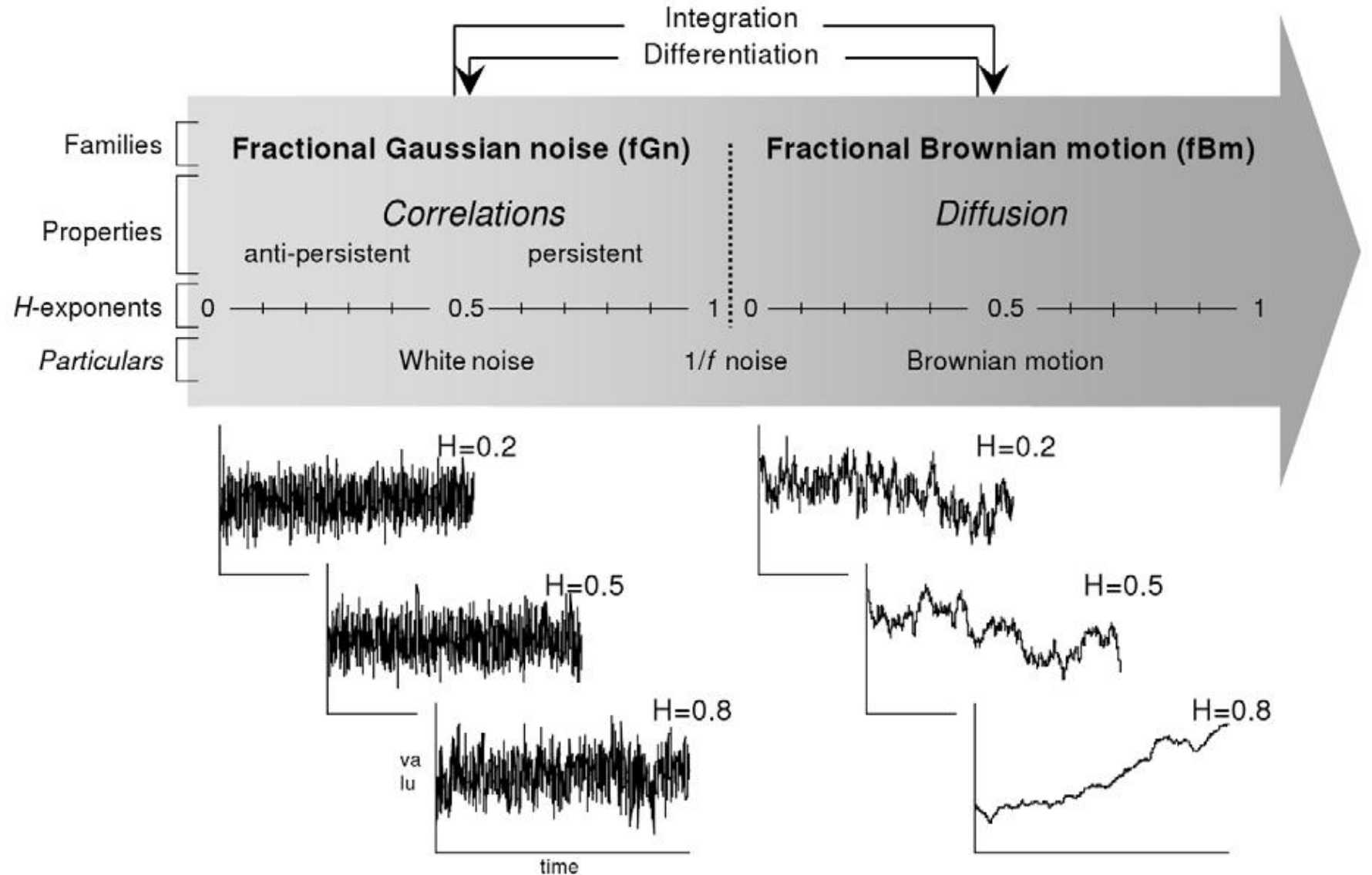
(c) Post-crash (2017-18)

Mukhia, K., Rai, A., Luwang, S. R., Nurujjaman, M., Majhi, S., & Hens, C. (2024)

3. Underlying Logic of Quantitative Strategies: Examples

Scientific Pattern

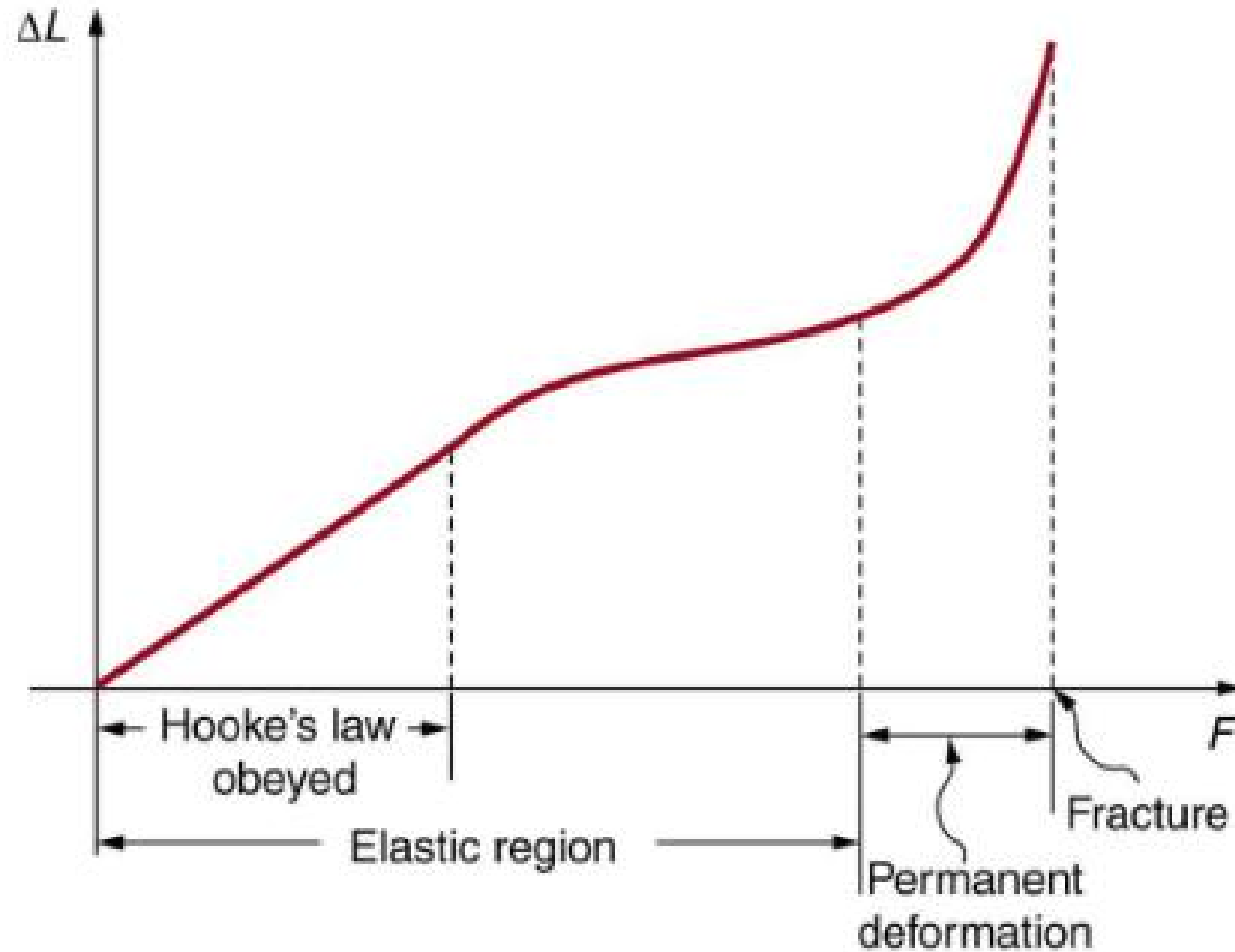
- Heat Equation
- Network Theory
- Chaos Theory
- Classical Mechanics



3. Underlying Logic of Quantitative Strategies: Examples

Scientific Pattern

- Heat Equation
- Network Theory
- Chaos Theory
- Classical Mechanics



source: x-Douglas College Physics 1107 Fall 2019 Custom Textbook

Appendix:

Core Logic Design of Trading Strategies		
Targeting Market		
Phenomena Description ☐	Theoretical Framework ☐	
Analytical Framework ☐	Theoretical Hypothesis ☐	
Phenomena Interpretation ☐	Theoretical Reasoning ☐	
Trading Opportunity		
Trading Operations		
Financial Instrument(s)		

Table 1: Core Logic Design of Trading Strategies

Design of Algorithmic Models	
Data Requirement	e.g., 5 min High, Close, Volume, news, 10K, etc.
Quantitative Model	
Strategy Type	Mean-reversion, momentum, event-driven, arbitrage, etc.
Trading Decision Triggers	Threshold, Cross, Decision Fusion Signals, etc.
Expected Return Level (1-5)	
Expected Risk Level (1-5)	
Expected Capacity Level (1-5)	
Expected Frequency Level (1-5)	
Parameters* (definition and possible ranges)	

Table 2: Design of Quantitative Models

Assignment 1 (Regular Track): Crypto Market Analysis

Lecture 4

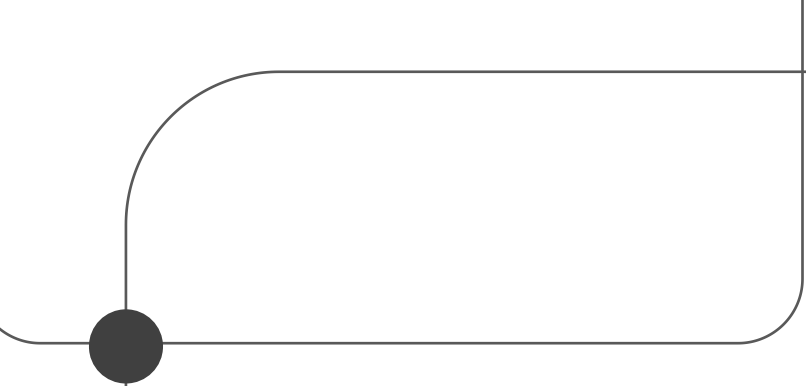
1.(30) **NO AI HERE!** Try to practice on your own mind and finish it before Lecture 5. That's a daily routine for quant researchers and traders.

Write an essay (200-500 words) about your trading idea. Illustrate your logic, pattern, model, trading rules and expected outcomes. This is the very first strategy draft: no detailed data, testing, or code is required.

Lecture 5

2.(70) **AI is fine here.** Reorganize your trading idea with Table 1 (Core Logic Design of Trading Strategies).

Originality and novelty in your trading idea is encouraged and will be considered in grading. However, note that a clear, well-reasoned, and critical analysis of an existing concept can be just as valuable as a completely novel idea.



Thank You

Hanchao Yang, PhD
hanchao.yang@duke.edu

